Protocol: I8B-MC-ITRV(b)
A Study to Evaluate the Pharmacokinetics and Glucodynamics of LY900014 compared

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Protocol I8B-MC-ITRV(b)
A Study to Evaluate the Pharmacokinetics and Glucodynamics of LY900014 compared to Humalog® in Patients with T1DM

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LY900014

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Table of Contents

A Study to Evaluate the Pharmacokinetics and Glucodynamics of LY900014 compared to Humalog® in Patients with T1DM

Section **Page**

Protocol I8B-MC-ITRV(b) A Study to Evaluate the Pharmacokinetics and Glucodynamics of LY900014 compared to Humalog® in Patients with T1DM 1
Table of Contents 2
1. Protocol Synopsis 8
2. Schedule of Activities 11
3. Introduction 17
3.1. Study Rationale 17
3.2. Background 17
3.3. Benefit/Risk Assessment 18
4. Objectives and Endpoints 20
5. Study Design 21
5.1. Overall Design 21
5.2. Number of Participants 22
5.3. End of Study Definition 23
5.4. Scientific Rationale for Study Design 23
5.5. Justification for Dose 23
6. Study Population 24
6.1. Inclusion Criteria 24
6.2. Exclusion Criteria 26
6.2.1. Additional Exclusion Criteria for Inpatient Dosing Days 28
6.3. Lifestyle and/or Dietary Requirements 28
6.3.1. Meals and Dietary Restrictions 28
6.3.2. Caffeine, Alcohol, and Tobacco 29
6.3.3. Activity 29
6.4. Screen Failures 29
7. Treatment 30
7.1. Treatment Administered 30
7.1.1. Packaging and Labeling of Investigational Medicinal Product 30
7.2. Method of Treatment Assignment 31

7.2.1. Selection and Timing of Doses 31
7.3. Blinding 31
7.4. Dose Modification 32
7.4.1. Special Treatment Considerations 32
7.5. Preparation/Handling/Storage/Accountability 32
7.6. Treatment Compliance 33
7.7. Concomitant Therapy 33
7.8. Treatment after the End of the Study 34
8. Discontinuation Criteria 35
8.1. Discontinuation from Study Treatment 35
8.1.1. Discontinuation of Inadvertently Enrolled Patients 35
8.2. Discontinuation from the Study 35
8.3. Patients Lost to Follow-up 36
9. Study Assessments and Procedures 37
9.1. Efficacy Assessments 37
9.2. Visit Description 37
9.2.1. Lead-In and Insulin Transition 37
9.2.2. Dose-Finding Assessment 38
9.2.3. Study Period CRU Visits 39
9.2.3.1. Run-In 39
9.2.3.2. Study MMTT 39
9.3. Adverse Events 39
9.3.1. Serious Adverse Events 40
9.3.1.1. Suspected Unexpected Serious Adverse Reactions 41
9.3.2. Complaint Handling 41
9.4. Treatment of Overdose 41
9.5. Safety 42
9.5.1. Physical Examination 42
9.5.2. Laboratory Tests 42
9.5.3. Vital Signs 42
9.5.4. Electrocardiograms 42
9.5.5. Other Tests 43
9.5.5.1. Hip and Waist Circumference 43
9.5.5.2. Body Weight and Height 43
9.5.6. Safety Monitoring 43
9.5.6.1. Hepatic Safety 43
9.5.6.2. Glucose Monitoring 44
9.5.6.3. Severe Hypoglycemia 45

9.5.6.4. Injection Site Assessments (Local Tolerability) 45
9.5.7. Self-Monitored Plasma Glucose during Outpatient Period 45
9.5.8. Treatment of Hyperglycemia and Hypoglycemia 46
9.6. Pharmacokinetics Samples 46
9.6.1. Bioanalysis 46
9.7. Pharmacodynamics 46
9.7.1. Glucodynamics Samples 47
9.7.1.1. Glucose Samples (Run-In Period) 47
9.7.1.2. Glucose Samples (MMTT) 47
9.7.2. Samples for Immunogenicity Assessments 47
9.8. Genetics 47
9.9. Biomarkers 48
9.10. Health Economics 48
10. Statistical Considerations and Data Analysis 49
10.1. Sample Size Determination 49
10.2. Populations for Analyses 49
10.2.1. Study Participant Disposition 49
10.2.2. Study Participant Characteristics 49
10.3. Statistical Analyses 49
10.3.1. Safety Analyses 50
10.3.1.1. Clinical Evaluation of Safety 50
10.3.1.2. Statistical Evaluation of Safety 50
10.3.2. Pharmacokinetic Analyses 50
10.3.2.1. Pharmacokinetic Parameter Estimation 50
10.3.2.2. Pharmacokinetic Statistical Inference 51
10.3.3. Glucodynamic Analyses 51
10.3.3.1. Glucodynamic Parameter Estimation 51
10.3.3.2. Glucodynamic Statistical Inference 52
10.3.4. Evaluation of Immunogenicity 52
10.3.5. Interim Analyses 52
11. References 53

List of Tables

Table **Page**

Table ITRV.1. Objectives and Endpoints 20

Table ITRV.2. Treatments Administered..... 30

List of Figures

Figure **Page**

Figure ITRV.1. Illustration of study design for Protocol I8B-MC-ITRV. 21

List of Appendices

Appendix **Page**

Appendix 1. Abbreviations and Definitions 54
Appendix 2. Clinical Laboratory Tests 58
Appendix 3. Study Governance, Regulatory and Ethical Considerations 59
Appendix 4. Hepatic Monitoring Tests for Treatment-Emergent Abnormality 62
Appendix 5. Blood Sampling Summary 63
Appendix 6. Protocol Amendment I8B-MC-ITRV(b) Summary: A Study to Evaluate the Pharmacokinetics and Glucodynamics of LY900014 compared to Humalog® in Patients with T1DM 64

1. Protocol Synopsis

Title of Study:
A Study to Evaluate the Pharmacokinetics and Glucodynamics of LY900014 compared to Humalog in Patients with T1DM

Rationale:
LY900014 is an ultra-rapid acting insulin lispro formulation with increased early absorption compared to commercially available insulin lispro formulation (Humalog®, Eli Lilly). LY900014 aims to mimic the physiological prandial insulin secretion pattern, which may more effectively, control postprandial glucose excursions and allow increased flexibility of the time of dosing relative to a meal.

The aim of this study is to compare the insulin lispro pharmacokinetic (PK), glucodynamic (GD), and tolerability profiles of insulin lispro following administration of either LY900014 or Humalog during a mixed meal tolerance test (MMTT) in patients with T1DM.

Objectives/Endpoints:

Objectives	Endpoints
Primary	
1. To evaluate the PK of insulin lispro following a single SC injection of LY900014 and Humalog in patients with T1DM	1. Early 50% t_{max} and AUC(0-30min)
Secondary	
1. To evaluate the effect of injection-to-meal timings (immediately before the start of meal, and 20 minutes following the start of the meal) on the GD response to LY900014 compared to Humalog, as measured by the MMTT	1. ΔAUC(0-2h), ΔAUC(0-5h)
2. To evaluate the tolerability of LY900014	2. AEs and hypoglycemic events

Abbreviations: AE = adverse event; AUC(0-30min) = area under the concentration versus time curve (AUC) from time zero to 30 minutes; ΔAUC = area under the baseline subtracted glucose concentration versus time curve; early 50% tmax = time to early half-maximal drug concentration; GD = glucodynamics; MMTT = mixed meal tolerance test; PK = pharmacokinetics; SC = subcutaneous; T1DM = type 1 diabetes mellitus

Summary of Study Design:

Study I8B-MC-ITRV is a Phase 1, randomized, patient and investigator blind, 2-treatment, 4-period crossover study to evaluate meal to dose-timing in patients with T1DM.

Treatment Arms and Planned Duration for an Individual patient:
Patients will be screened over a 14 day period prior to start of an approximate 2-week lead-in/insulin transition period. Patients will then participate in a dose-finding assessment (Day -1 of Period 1) and subsequently

randomised to 1 of 4 treatment sequences according to the actual randomization table provided to the site. In each study period, patients will undergo an MMTT. Each patient will receive a single subcutaneous injection of either LY900014 or Humalog either immediately before the test meal or 20 minutes following the start of the test meal. All treatment procedures will be inpatient stays of approximately 2 days per period and require a maximum of 6 weeks to complete 4 periods. Each study dosing will be separated by a minimum of 21 hours and may occur on consecutive visits, however patient inclusion is subject to required screening criteria. The follow-up or early discontinuation visit should occur at least 14 days after the last dose of the study drug.

Treatment:

LY900014: Single individualized SC dose per assessment period

Humalog: Single individualized SC dose per assessment period

Number of Patients:
Up to 36 patients may be enrolled to ensure that at least 30 patients will complete the study. Thirty completing patients will provide greater than 95% power to demonstrate a 2-fold increase in the serum insulin lispro AUC(0-30min) between LY900014 and Humalog when both are given immediately before meals.

Statistical Analysis:
Primary statistical analyses of PK and GD parameters will be conducted on the set of patients who complete all treatment periods. Supportive analyses may be done on the key parameters for the patients who complete at least the first period of treatment. Safety analyses will be conducted on the set of patients receiving at least 1 dose of the study drug to which they are randomized, regardless of whether or not they completed all protocol requirements.

Unless otherwise noted, all tests of treatment effects will be conducted at a 2-sided alpha level of 0.05, and confidence intervals (CIs) will be calculated at 95%, 2-sided.

Safety: All investigational product and protocol procedure AEs will be listed, and if the frequency of events allows, safety data will be summarized using descriptive methodology. Safety parameters will be listed and summarized using standard descriptive statistics.

Pharmacokinetic: Log transformed AUCs, Cmax, CL/F and Vz/F for insulin lispro will be evaluated to estimate geometric means, ratios of geometric means of insulin lispro within LY900014 to Humalog, and the corresponding 95% CIs of the ratios using the mixed-effects model that includes treatment and period as fixed effects and patient as a random effect.

The same model without log transformation will be used for the analysis of the PK time parameters (early 50% t_{max}, late 50% t_{max}, t_{max}, and $t_{1/2}$). Least-squares means (LSmeans), treatment differences in LSmeans, and the corresponding 95% CIs for the treatment differences will be estimated from the model. The p-value on the difference between LS means will be used to determine statistical significance. The treatment ratios and 95% CIs for the ratios will be calculated using Fieller's theorem.

Glucodynamic: Data will be analyzed for the patients during each MMTT. The change from baseline values (the average of -30, -15, and 0 minutes) represented as the 0-hour time point following the start of the MMTT for each patient will be calculated. The area under the baseline subtracted glucose concentration versus time curve from time 0 to 2hr (ΔAUC[0-2h]) and area under the baseline subtracted glucose concentration versus time curve from time 0 to 5hr post meal (ΔAUC[0-5h]) will be calculated. In addition, the change from baseline maximum glucose

observed during the 5 hours postmeal and change from baseline 1 hour glucose and 2 hour glucose post the start of the meal will be calculated. Other partial ΔAUCs may be calculated as deemed appropriate.

Summary statistics will be presented by treatment and by timing of dose (all 4 combinations of treatment and timing of dose). The GD parameters on the original scale will be analyzed using the mixed-effects model that includes treatment, timing of dose, treatment-by-timing of dose interaction, and period as fixed effects and patient as a random effect.

The p-value on the difference between LS means will be used to determine statistical significance and the corresponding 95% CIs for the LS Mean ratios from Fieller's theorem will be presented.

2. Schedule of Activities

Study Schedule Protocol I8B-MC-ITRV

	Sign Informed Consent	Screening	Lead-in	Study Period (Periods 1 to 4)		Follow-up/Early discontinuation	Comments
Procedure		Days -30 to -17	14 days prior to Day -2	Day -1	Day 1	≥ 14 days after discharge	
Informed Consent	X						At least 1 day before screening procedures. Screening procedures should take place no later than 28 days after signing the informed consent. (see Section 9.5.2).
Medical History and physical examination		X					
Hip and Waist Circumference				X			Period 1 only
Height		X					At screening only
Weight		X		X			Period 1 only
Vital signs: blood pressure and pulse rate		X			Predose, 30 and 120 minutes postdose	X	Vital signs will be taken while patients are supine at screening and semi-supine on Day 1.
12-lead ECG		X			Predose (Period 1)	X	Single ECGs will be collected for safety.
Clinical laboratory tests		X			Predose for Period 1 only	X	Fasting laboratory test for screening and follow-up. Screening laboratory tests will be analyzed at a local laboratory (see Section 9.5.2).

	Sign Informed Consent	Screening	Lead-in	Study Period (Periods 1 to 4)		Follow-up/Early discontinuation	Comments
Procedure		Days -30 to -17	14 days prior to Day -2	Day -1	Day 1	≥ 14 days after discharge	
Lead-in/insulin transition activities			X (Period 1 only)				Once patients have completed screening procedures, they will switch from prescribed short-acting and basal insulin to site-provided Humalog and insulin glargine respectively. Patients receive general diabetes training. A patient diary will be provided for recording dosing and other required information (Section 9.2.1) Lead-in period may be extended by 2 days (e.g., 16 days prior to Day -2) as needed
Patient Admission to CRU				X			Admitted to CRU evening Day -2 or early morning Day -1 (Period 1 only). For Periods 2 – 4, patients may be admitted evening of Day -1.
Dose-finding				X (Period 1 only)			Following insulin transition, patients will undergo dose-finding with Humalog and a test meal. To take place between 07:00 and 11:00. Blood glucose concentrations will be monitored every 20 minutes for 5 hours postmeal (see Section 9.2.2). See Section 6.2.1 for rescheduling.

	Sign Informed Consent	Screening	Lead-in	Study Period (Periods 1 to 4)		Follow-up/Early discontinuation	Comments
Procedure		Days -30 to -17	14 days prior to Day -2	Day -1	Day 1	≥ 14 days after discharge	
Randomization				X			To take place before run-in of Period 1 Day 1
Pregnancy test		X		X		X	Serum pregnancy test will be performed at screening and urine pregnancy tests will be performed at admission for Period 1 and at follow-up (see Appendix 2)
Standard dinner				X			Approximately 19:00
Medical Assessment				X	Predose and before discharge from CRU	X	Medical assessment includes medical review and targeted examination, and as appropriate review of concomitant medication, patient diary and MMTT exclusion criteria (see Section 6.2.1)
Glucose stabilization/run-in					X		From approximately 7 hours to 30 minutes before dosing: infusion of glucose (dextrose solution) or insulin glulisine to achieve a target blood glucose concentration of 135 ±15mg/dL (7.5 ±0.8 mmol/L). Blood glucose concentrations will be monitored at a minimum of 30-minute intervals (see Section 9.2.3.1 and 9.7.1.1). See Section 6.2.1 for rescheduling.

	Sign Informed Consent	Screening	Lead-in	Study Period (Periods 1 to 4)		Follow-up and Early discontinuation	Comments
Procedure		Days -30 to -17	14 days prior to Day -2	Day -1	Day 1	≥ 14 days after discharge	
Blood glucose sampling (MMTT)					-30, -15, 0 minutes pre-meal, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 135, 150, 165, 180, 195, 210, 225, 240, 300 minutes post-meal		0-minute time point sample to be taken at the start of the meal.
Mixed Meal Tolerance Test (MMTT)					X		Standardized liquid test meal will be administered at approximately 08:00 (with allowance to 11:00) and should be consumed within 15 minutes. (See Section 9.2.3.2). Study periods to be separated by a minimum 21 hour washout
Study drug administration					X		Either immediately before meal or 20 minutes after start of meal according to randomization schedule: Study drug will be administered at approximately the same time on Day 1 of Periods 1-4. There will be at least 21 hours between study doses.
Injection site local tolerability assessments					0, 20, and 60 minutes postdose		Assessments of injection site local tolerability will occur immediately following the injection

	Sign Informed Consent	Screening	Lead-in	Study Period (Periods 1 to 4)		Follow-up and Early discontinuation	Comments
Procedure		Days -30 to -17	14 days prior to Day -2	Day -1	Day 1	≥ 14 days after discharge	
Insulin lispro PK sampling					0 (predose), 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 70, 90, 120, 150,180, 240, 300, 360, and 420 minutes postdose		Sampling times are relative to the time of study drug administration in each period
Pharmacogenetics sample					Predose for Period 1 only		Refer to sample collection instructions provided by the sponsor
Immunogenicity sample			Pre-insulin switch		Predose for Periods 1 and 3	X	Additional samples may be collected if the investigator considers there is a possibility that an AE is immunologically indicated.
Discharge from CRU					X		Patients may be discharged after all study procedures are completed

Study Schedule Protocol I8B-MC-ITRV

Abbreviations: CRU = clinical research unit; ECG = electrocardiogram; ED = early discontinuation; min = minutes; MMTT = mixed meal tolerance test; PK = pharmacokinetics

Note: The site should schedule activities as appropriate. In cases where several study procedures are scheduled at the same time, the order of priority will be as follows: PK samples including blood sampling for blood glucose and laboratory samples per protocol nominal times. ECGs and vital sign measurements should be scheduled before but as close as possible to the PK sampling times. Injection-site assessments can be done after PK sampling.

3. Introduction

3.1. Study Rationale

A prandial insulin with faster-on and/or faster-off characteristics might reduce glycemic excursions and the incidence of postprandial hypoglycemia compared to currently available fast-acting insulin analogs. LY900014 is an ultra-rapid-acting insulin lispro formulation that has shown an increased early absorption compared to commercially available insulin lispro (Humalog®; Eli Lilly). LY900014 aims to closely mimic the physiological prandial insulin secretion pattern, which may more effectively control postprandial glucose excursions and allow increased flexibility of the time of dosing relative to a meal.

This study aims to compare PK and PD profiles as well as the improved action of insulin lispro from LY900014 on the control of postprandial blood glucose in comparison to Humalog in patients with T1DM. The improved action of insulin lispro will be assessed by using standardized mixed-meal tolerance test (MMTTs). Furthermore, this study will explore the postprandial glucose profiles with LY900014 and Humalog injected at two different meal-to-dose time intervals either immediately before or 20 minutes following the start of the test meal, to optimize future use of this investigational medicinal product (IMP).

3.2. Background

The insulin analog insulin lispro (Humalog) has been shown to be absorbed more quickly than regular human insulin (Humalog package insert, 2015). In healthy volunteers given subcutaneous (SC) doses of insulin lispro ranging from 0.1 to 0.4 units (U)/kg, peak serum levels were seen 30 to 90 minutes after dosing (Humalog package insert, 2015). However, the general consensus is that rapid-acting insulin is still not rapid enough to match carbohydrate absorption profiles, which limits efficacy and dosing flexibility. An ultra-rapid-acting prandial insulin would shift the PK/GD of insulin analogs so that they have an even faster onset to better match carbohydrate absorption and also allow greater flexibility in the time of dosing relative to meals.

LY900014 represents a new formulation that contains insulin lispro, treprostinil, citrate and other excipients. This formulation involves the novel use of a microdose of treprostinil ([redacted]C) as an excipient to enhance the absorption of insulin lispro by local vasodilatation rather than as an active pharmaceutical ingredient to elicit a systemic effect. Treprostinil is a prostacyclin analogue, administered either through inhalation ([redacted]C), as an intravenous (IV) infusion or as a continuous SC administration for the treatment of symptomatic pulmonary arterial hypertension (PAH) and has been approved in the US since 2002 and in Germany since 2006 (AMIS database [WWW]). Sodium citrate, an excipient that speeds insulin absorption (at least in part by enhancing vascular permeability), is also included in the formulation to further enhance the absorption of insulin lispro. Each of the other excipients (such as magnesium chloride) in the LY900014 formulation is listed in the US Food and Drug Administration (FDA)'s Generally Recognized as Safe Food Additives database and in the FDA's Inactive Ingredients in Approved Drugs database. Furthermore, the excipient concentration in LY900014 is within the limits identified for approved drug products in the FDA Inactive Ingredients in Approved Drugs database.

To date, LY900014 has been administered to approximately 89 patients with diabetes (60 T1DM and 29 T2DM) across 3 Phase 1b studies and approximately 107 healthy volunteers across these Phase 1 studies. The total insulin lispro exposure was similar between LY900014 and Humalog; however LY900014 demonstrated a faster and earlier insulin lispro absorption compared to Humalog. Additionally, through the use of multiple daily injections of LY900014 for up to 2 weeks in patients with either T1DM or T2DM, it was found that LY900014 was well tolerated. There were no serious adverse events (SAEs) related to study treatment or discontinuations from the studies because of a drug-related adverse event (AE). Small numbers of treatment-emergent adverse events were reported, and there were no notable increases in these events in relation to any of the LY900014 formulations compared to those in relation to Humalog. There were no reported cases of severe hypoglycemia. Additionally, there was no reported incidences of local or systemic allergic reactions.

3.3. Benefit/Risk Assessment

This study will not offer any direct benefits to the patients participating in the study. The data from previous studies in healthy patients and patients with T1DM and T2DM have shown that LY900014 was well tolerated and the adverse drug reactions are in keeping with those reported for Humalog.

Potential risks associated with LY900014, derived from the known risks of insulin lispro (Humalog), are hypoglycemia, hypersensitivity reactions (localized allergy and/or systemic allergy), undesirable effects at the injection site (injection-site reactions and lipodystrophy), and peripheral edema (Humalog package insert, 2015).

Notably, across all doses in the studies that have evaluated treprostinil (CCI) as a local vasodilator with or without insulin lispro, there was no clinically significant increase in those AEs associated with systemic absorption of treprostinil, as described in the CCI package insert (2014). The exposures of treprostinil in LY900014 in this study are expected to be undetectable compared to those observed in the dose ranges previously explored with SC bolus administration of treprostinil and are expected to be substantially lower than those observed in the treatment of PAH. No known potential risks are associated with the use of small amounts of treprostinil in the LY900014 formulation.

In preclinical safety pharmacology and toxicity studies, or clinical pharmacology studies involving LY900014 or treprostinil alone, other than known risks associated with Humalog and CCI , no additional risks were identified. Additionally, local and systemic toxicity profiles of Humalog and CCI do not suggest the potential for additive or synergistic toxicity

The study includes inpatient procedures during which participants will be continuously monitored. Patients will be without food intake from the start of the MMTT to completion of blood collection (approximately 420 minutes) unless required to treat hypoglycemia, (see Section 9.5.6.2 for definition of hypoglycemia and Section 9.5.8 for treatment guidelines). If a patient experiences elevated blood glucose following food intake for more than 1 hour, insulin glulisine will be administered intravenously (see Section 9.5.8 for treatment guidelines). Patients will maintain their basal insulin regimen of site-provided insulin glargine during the entire study

including the MMTT days unless safety issues arise; in this case, the investigator will discuss a change of the basal insulin regimen with the sponsor clinical pharmacologist (CP) and implement this change, if necessary, to prevent any medical problems.

More information about the known and expected benefits, risks, serious adverse events (SAEs) and reasonably anticipated adverse events (AEs) of LY900014 are to be found in the Investigator's Brochure (IB).

4. Objectives and Endpoints

Table ITRV.1 shows the objectives and endpoints of the study.

Table ITRV.1. Objectives and Endpoints

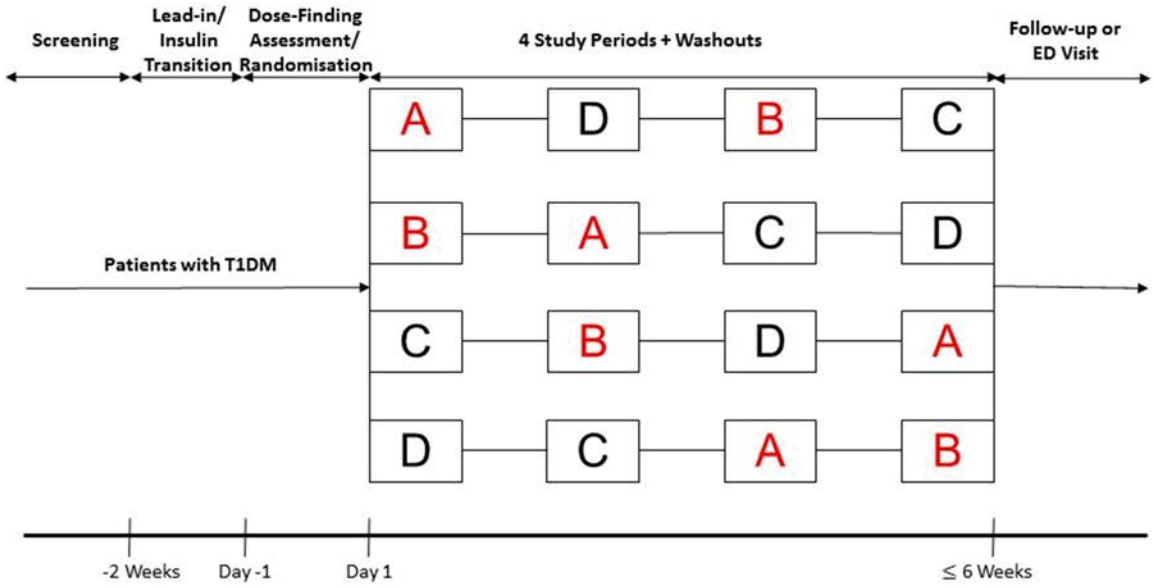
Objectives	Endpoints
Primary	
1. To evaluate the PK of insulin lispro following a single SC injection of LY900014 and Humalog in patients with T1DM	1. Early 50% t_{max} and AUC(0-30min)
Secondary	
1. To evaluate the effect of injection-to-meal timings (immediately before the start of meal, and 20 minutes following the start of the meal) on the GD response to LY900014 compared to Humalog, as measured by the MMTT	1. ΔAUC(0-2hr), ΔAUC(0-5hr)
2. To evaluate the tolerability of LY900014	2. AEs and hypoglycemic events
Exploratory	
1. Explore the formation of anti-drug antibodies to insulin lispro	1. Anti-insulin lispro antibodies

Abbreviations: AE = adverse event; AUC(0-30min) = area under the concentration versus time curve (AUC) from time zero to 30 minutes; ΔAUC = area under the baseline subtracted glucose concentration versus time curve; early 50% tmax = time to early half-maximal drug concentration; GD = glucodynamics; MMTT = mixed meal tolerance test; PK = pharmacokinetics; SC = subcutaneous; T1DM = Type 1 diabetes mellitus.

5. Study Design

5.1. Overall Design

ITRV is a Phase 1, patient- and investigator-blind, randomized, 2-treatment, 4-period crossover study in patients with T1DM to evaluate the insulin lispro PK and GD characteristics of LY900014 compared to Humalog following single SC injections administered at 2 different injection-to-mealtime intervals; either immediately before or 20 minutes following the start of the test meal. Study ITRV may be conducted at 1 or more clinical research units (CRUs).



Treatment	LY900014 or Humalog	Dose Timing Relative to Meal (minute)*
A	LY900014	0
B	LY900014	+20
C	Humalog	0
D	Humalog	+20

* Time 0 is immediately before test meal

Abbreviations: ED = early discontinuation; T1DM = type 1 diabetes mellitus

Figure ITRV.1. Illustration of study design for Protocol I8B-MC-ITRV.

Patients will be required to attend the CRU on at least 8 occasions (no more than 10 occasions if dose-finding and/or one MMTT rescheduled) as noted in the Study Schedule (see Section 2):

- Informed consent
- Screening visit
- Lead-in and insulin transition period (see Section 9.2.1)

- 4 inpatient CRU study visits (including dose finding assessment [see Section 9.2.2])
- A follow-up visit

Eligible patients who have satisfied the entry criteria and completed all screening procedures will return to the CRU at least 2 weeks prior to Period 1 Day -2 to begin a lead-in and insulin transition period. When patients visit the CRU to begin the lead-in period, patients will receive instruction on general diabetes education including measurement of self-monitored plasma glucose (SMPG) and receive instruction on the insulin transition (see Section 9.2.1). During the insulin transition period, patients will transition from their current basal insulin therapy to site-provided insulin glargine and Humalog [redacted] (see Section 9.2.1). Following insulin transition, patients will return to the CRU for the dose-finding assessment (see Section 9.2.2) on either the evening of Day -2, Period 1 or early morning of Day -1, Period 1. The dose-finding assessment will use Humalog administered immediately before a standardized liquid test meal (MMTT) to inform the dose of LY900014 and Humalog to be used during subsequent study MMTT assessments. Upon completion of the dose-finding assessment, patients will be randomised to 1 of 4 treatment sequences. Prior to the MMTT in each study period, patients will undergo a run-in period (see Section 9.2.3.1) to achieve a predetermined glucose target of 135 ±15 mg/dL (7.5 ±0.8 mmol/L). Once the glucose target has been attained, patients will proceed with the MMTT in which a single individualized SC dose of LY900014 or Humalog will be administered either immediately before the test meal or 20 minutes following the start of the test meal (see Section 9.2.3.2).

Doses of LY900014 and Humalog during the 4 study periods will be separated by a minimum washout of approximately 21 hours. The maximum duration allowed for all 4 periods is approximately 6 weeks. Patients will continue using site-provided Humalog and insulin glargine during the washout periods between MMTTs (see Section 7.7 for use of concomitant medication and basal insulin). During the washout periods, patients will be instructed to perform regular blood glucose monitoring (see Section 9.5.7).

Following completion of all study procedures in each Period, patients may be offered a meal and discharged from the CRU at the discretion of the principal investigator.

Study governance considerations are described in detail in Appendix 3.

5.2. Number of Participants

Up to 36 patients may be enrolled so that approximately 30 patients complete the study. For purposes of this study, a patient completes the study when all scheduled procedures shown in the Schedule of Activities have been finished. Patients who drop out may be replaced, and the replacement patient will adopt all 4 assigned crossover treatments of the original patient's randomization schedule.

5.3. End of Study Definition

End of the study is the date of the last visit or last scheduled procedure shown in the Schedule of Activities (Section 2) for the last patient.

5.4. Scientific Rationale for Study Design

The study is a 4-period crossover design to reduce the variability of insulin PK and GD, as each patient will act as his/her own control. The total number of patients needed with a crossover design is less than the number needed with a parallel group design. A maximum duration of approximately 6 weeks is allowed for patients to complete all 4 assigned periods to minimize the risk of insulin resistance/changes in mean glycemic control during the study.

Randomization and blinding are used to avoid bias introduced through an association between allocation order of investigational medicinal product (IMP) and patient characteristics. The Lilly clinical pharmacologist (CP)/Lilly study team will be unblinded.

In each study period patients will undergo a run-in period before the MMTT using a variable insulin and glucose IV infusion. This run-in period will allow for improved comparability of the postprandial glucose response to a mixed meal after treatment with LY900014 and Humalog administered either immediately before the start of the meal, or 20 minutes following the start of the test meal. This run-in aims to achieve similar preprandial glucose levels for all patients before the start of the test meal and thereby reduces the variability of the postprandial glucose response. Insulin glulisine has been chosen for the IV optimization of blood glucose during the run-in because insulin glulisine does not cross-react with the insulin lispro–specific assay used for the PK analysis.

Under this design, if two periods occur on consecutive days, the interval between the last bolus on the first day and the first bolus on the second day is much longer compared to the length of time that the treatment (LY900014 or Humalog) lasts in the bloodstream; therefore, no carryover effect is assumed. This enables PK and GD data from the breakfast meal tests of each period to be analyzed independently and separately.

A minimum of approximately 21 hours washout between the test meals allows for a complete washout of study drug administered with the MMTT and glucose response and prevents carry-over effects.

5.5. Justification for Dose

The bolus dose of insulin lispro (LY900014 or Humalog) will be individualized per patient to cover the carbohydrate content in this standardized liquid test meal. This dose of insulin is reflective of a clinically relevant, individualized insulin dosing, similar to how patients would determine the insulin dose for the carbohydrate content in a meal. For each patient, the individualized prandial insulin lispro dose in LY900014 and Humalog for each test meal must be kept identical throughout the crossover periods.

6. Study Population

Eligibility of patients for study enrollment will be based on the results of screening medical history, physical examination, vital signs, clinical laboratory tests, and electrocardiogram (ECG).

The nature of any conditions present at the time of the physical examination and any preexisting conditions will be documented.

Screening may occur over a 14 day period prior to the lead-in site visit. Patients who are not entered within this time period, may be subjected to an additional medical assessment and/or clinical measurements to confirm their eligibility.

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, are not permitted.

6.1. Inclusion Criteria

Patients are eligible for inclusion in the study only if they meet all of the following criteria at screening and/or enrollment:

[1] are male or female patients with T1DM for at least 1 year. A diagnosis of T1DM is based on medical history with a fasting C-peptide $\leq$0.30 nmol/L.

a. Male patients:

i. No male contraception required except in compliance with specific local government requirements

b. Female patients:

i. Women of child-bearing potential who are abstinent (if this is complete abstinence, as their preferred and usual lifestyle) or in a same sex relationship (as part of their preferred and usual lifestyle) must agree to either remain abstinent or stay in a same sex relationship without sexual relationships with males.

ii. Otherwise, women of child-bearing potential participating must agree to use one highly effective method of contraception until discharge from final treatment period.

1. Women of child bearing potential must test negative for pregnancy prior to initiation of treatment as indicated by a negative serum pregnancy test at the screening visit followed by a negative urine pregnancy test within 24 hours prior to exposure in Period 1.

2. A highly effective method of contraception includes a combined (estrogen and progestogen containing) or progestogen-only hormonal contraception administered orally, intravaginally, or transdermally and is associated with inhibition of ovulation. Alternatively, patients may use either an intrauterine device, intrauterine hormone-releasing system, bilateral tubal occlusion, or the partner should have been vasectomised.

iii. Women not of childbearing potential may participate and include those who are:

1. infertile due to surgical sterilization (hysterectomy, bilateral oophorectomy or bilateral salpingectomy), congenital anomaly such as mullerian agenesis; or

2. post-menopausal – defined as a woman being amenorrheic for more than 1 year without an alternative medical cause and a serum follicle-stimulating hormone (FSH) level compatible with postmenopausal status. A FSH level in the post-menopausal range may be used to confirm a post-menopausal state in women not using hormonal contraception or hormonal replacement therapy.

[2] are aged 18 to 70 years inclusive at the time of screening.

[3] have a body mass index (BMI) of 18.5 kg/m^2 to 35.0 kg/m^2, both inclusive, at screening.

[4] have clinical laboratory test results within normal reference range for the population or investigative site, or results with acceptable deviations that are judged to be not clinically significant by the investigator.

[5] have venous access sufficient to allow for glucose infusion and blood sampling procedures as per protocol.

[6] are reliable and willing to make themselves available for the duration of the study and are willing to follow study procedures.

[7] are able and willing to give signed informed consent approved by Lilly and the ethical review board (ERB) governing the site.

[8] have a glycated hemoglobin (HbA_1c) ≤9.0% at screening.

[9] have had no episodes of severe hypoglycaemia in the last 6 months (see Section 9.5.6.3).

[10] are on stable prandial insulin and basal insulin (neutral protamine Hagedorn [NPH] insulin, insulin glargine or insulin detemir) for at least 3 months before screening with a total insulin dose demand of ≤1.5 U/kg/day

6.2. Exclusion Criteria

Patients will be excluded from study enrollment if they meet any of the following criteria at screening and/or enrollment:

[11] are investigative site personnel directly affiliated with this study and their immediate families. Immediate family is defined as a spouse, biological or legal guardian, child, or sibling.

[12] are Lilly employees or employees of the investigational site/CRU.

[13] are currently enrolled in a clinical study involving an investigational product or any other type of medical research judged not to be scientifically or medically compatible with this study.

[14] have participated within the last 30 days in a clinical trial involving an investigational product. If the previous investigational product has a long half-life, 3 months or 5 half-lives (whichever is longer) should have passed.

[15] have previously completed or withdrawn from this study.

[16] have an abnormality in the 12-lead ECG that, in the opinion of the investigator, increases the risks associated with participating in the study.

[17] have a supine blood pressure at screening outside the range of 90 to 160 mmHg for systolic or 50 to 100 mmHg for diastolic (1 repeat is allowed) as determined by the investigator, or results with unacceptable deviations that are judged by the investigator to be clinically significant for the population, or have a heart rate outside the range of 50 to 90 beats/minute.

[18] have a history or current cardiovascular, respiratory, hepatic, renal, gastrointestinal, endocrine (apart from T1DM), hematological, or neurological disorders capable of significantly altering the absorption, metabolism, or elimination of drugs; of constituting a risk when taking the investigational product; or of interfering with the interpretation of data.

[19] have known or ongoing psychiatric disorders.

[20] regularly use known drugs of abuse and/or show positive findings on urinary drug screening.

[21] show evidence of an acute infection with fever or infectious disease at the time of study entry.

[22] show evidence of human immunodeficiency virus (HIV) infection and/or positive human HIV antibodies.

[23] show evidence of hepatitis C and/or positive hepatitis C antibody (the presence of hepatitis C antibodies in the setting of normal liver function tests and a negative hepatitis C polymerase chain reaction are not an exclusion).

[24] show evidence of hepatitis B and/or positive hepatitis B surface antigen (the presence of antibodies to the hepatitis B surface antigen is not an exclusion).

[25] are women who are pregnant or lactating.

[26] have significant lipohypertrophy in the target abdominal injection area as judged by the investigator.

[27] have a history of renal impairment (exclusion only if glomerular filtration rate [estimated GFR] <60 mL/minute/1.73 m^2 [GFR is estimated according to the Chronic Kidney Disease Epidemiology Collaboration creatinine equation], or have a serum creatinine level ≥126µmol/L (male) or ≥111 µmol/L (female).

[28] have a history of deep vein thrombosis of the leg or repeated episodes of deep leg vein thrombosis in first-degree relatives (parents, siblings, or children) as judged by the investigator.

[29] have proliferative retinopathy or maculopathy and/or severe neuropathy; in particular, autonomic neuropathy as judged by the investigator based on a recent (<1.5 years) ophthalmologic examination.

[30] are currently a smoker as defined as more than 5 cigarettes (or nicotine equivalent) per day.

[31] have donated blood of more than 450 mL or more in the last 3 months or provided any blood donation within the last month before screening.

[32] have a significant history of alcoholism or drug abuse as judged by the investigator or consuming more than 24 g of alcohol per day for men, or more than 12 g of alcohol per day for women (1 unit of alcohol is defined as 10 mL [8 g] of pure alcohol).

[33] are unwilling to comply with the dietary requirements/restrictions during the study: (i) comply with the fasting requirements of the study, (ii) consume only the meals/snacks provided during the inpatient visits.

[34] have, except for current regimen of insulin therapy and concomitant medication(s) (for example, antihypertensive medication, lipid-lowering agent, thyroid hormone replacement medication, hormonal contraception, hormonal replacement therapy), regular use of or intended use of any over-the-counter or prescription medications or nutritional supplements that treat hyperglycaemia or insulin resistance or that promote weight loss within 14 days before dosing.

[35] are receiving chronic (lasting longer than 14 consecutive days) systemic or inhaled glucocorticoid therapy (excluding topical, intra-articular, and intraocular preparations), or have received such therapy within 4 weeks before screening.

[36] are treated with a CSII (insulin pump)

[37] have regular use of or intended use of known inducers or inhibitors of cytochrome P450 [CYP]2C8 or niacin. Patients taking these medications before study enrollment will be eligible with an appropriate washout period which will be evaluated by the investigator in consultation with the Lilly CP

[38] have known allergies to treprostinil (CCI), insulin lispro, insulin glulisine, insulin glargine, related compounds, or any components of the formulation, or a history of significant atopy.

[39] any significant changes in insulin regimen and/or unstable blood glucose control within the past 3 months prior to screening as assessed by the investigator.

[40] in the opinion of the investigator or sponsor, are unsuitable for inclusion in the study.

6.2.1. Additional Exclusion Criteria for Inpatient Dosing Days

Patients who fulfil 1 or more inpatient dosing-day exclusion criteria will be excluded from study drug administration for that MMTT day. A single inpatient treatment period can be rescheduled 1 to 14 days later. Each treatment period can only be rescheduled once.

The following exclusion criteria apply to the day ***before*** each MMTT day:

[41] positive drug screen

[42] positive pregnancy test administered on Day -1 of Period 1

[43] any medical condition or AE that could interfere with glucose metabolism, as judged by the investigator

[44] any use of prescription or non-prescription medication according to Exclusion Criteria [34] and [35]

[45] hypoglycemia during the treatment period and less than 24 hours before dosing that poses a significant risk to patient safety, as judged by the investigator

[46] injection of a bolus of more than 6 U of a short-acting insulin (Humalog) between 7 and 12 hours before test meal

6.3. Lifestyle and/or Dietary Requirements

Throughout the study, patients may undergo medical assessments and review of compliance with requirements before continuing in the study.

6.3.1. Meals and Dietary Restrictions

Patients will be fasted (except for water) for at least 10 hours before each test meal and 8 hours prior to screening and follow-up assessments (see Section 9.5.8 for guidelines on hypoglycemia treatment). The patient will not be allowed to consume water for 2 hours after the MMTT assessment begins; however, water may be consumed freely 2 hours postmeal.

While resident in the CRU, patients may not consume any food or caloric drinks other than that provided by the CRU. When not resident in the CRU, patients may resume their regular diet and should not make major changes to the dietary habits during the study.

Following completion of study procedures, patient will be offered a meal. Site will record dose and time of prandial Humalog administered with end-of-study meal.

6.3.2. Caffeine, Alcohol, and Tobacco

Patients should refrain from caffeine-containing food/beverages (for example, cola, chocolate drinks, tea, coffee, energy drinks containing methylxanthine [caffeine, theophylline, or theobromine]) for at least 12 hours before each test meal and throughout the duration of each CRU visit.

No alcohol will be allowed at least 24 hours before each CRU admission (Day –1) and throughout the duration of each CRU visit. Between CRU visits, daily alcohol should not exceed 3 units for males and 2 units for females (see Section 6.2 #32 for unit definition).

No cigarette smoking will be permitted during the study.

6.3.3. Activity

Patients will be encouraged to maintain their regular exercise and insulin regimen adaptation related to exercise during the outpatient period; however, they should not undertake vigorous or prolonged exercise at least 24 hours before each dosing day at the CRU. Movement will be restricted to retain the integrity of connections to infusion(s) and the study procedures.

6.4. Screen Failures

Individuals who do not meet the criteria for participation in this study may not be rescreened.